-: Physics :-

MCQs::

1. C
2. ~~A~~B
3. C
4. B
5. B
6. B
7. C
8. B
9. C
10. C
11. C
12. B

Q: 2(i)

A: 2(i)

North : 3m/s

East : 4m/s

magnitude :

$$\sqrt{(3)^2 + (4)^2}$$

$$\sqrt{(9) + 16}$$

$$\sqrt{25}$$

5m/s

velocity is a vector quantity because it has both magnitude and direction.

Q:2(ii)
A:2(ii)

(a) Acceleration at first stage :-

$v_i = 0\ m/s$
$v_f = 20\ m/s$
$t = 10s$

$$a = \frac{v_f - v_i}{t} \Rightarrow \frac{20-0}{10}$$

$$= \frac{2\cancel{0}}{1\cancel{0}} \Rightarrow 2\ m/s^2$$

(b) Distance :-

$$S = v_i \times t + \frac{1}{2} a(t)^2$$

$$S = (0 \times 10) + \frac{1}{2}(2)(10)^2$$

$$S = 0 + \frac{1}{\cancel{2}}\cancel{2}(10)^2$$

$$S = 0 + 100$$

$$S = 100m$$

~~speeds Q.2(~~

(ii) constant ~~acceleration~~ :

speed = 20 m/s

$\cancel{v_i = 20}$

$\cancel{v_f = 0}$

$\cancel{t = 25}$

$S_2 = v \times t$

$= (20)(25)$

$= 500m$

$\cancel{a = \frac{0 - 20}{25}}$

$\cancel{\frac{-20}{25}}$

Total ~~s~~ distance :- 500 + 100

600 m.

Q:2(iii)

A:2(iii)

(a) Gravitational field strength is the gravitation force acting on a unit mass placed in a gravitational field.

formula :- $g = \frac{F}{m}$

(b) Given :- m = 8 Kg

gravitational field = ~~8~~3.7 N

$W = mg$

$W = 8 \times 3.7$

$W = 29.6 N$

Q: 2(i)
A: 3(i)

(a) 1st conditions of equilibrium:-

When the vector sum of all forces acting on the body is zero then the first condition of equilibrium is satisfied.

Mathematically, if F_{net} is the sum of forces $\vec{F}_1, \vec{F}_2, \vec{F}_3 \dots \vec{F}_n$ then:-

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots + F_n = 0$$

$$\Sigma F = 0$$

2nd conditions of equilibrium:-

When the vector sum of all torques acting on the body is zero then the second condition of equilibrium is satisfied.

Mathematically, if $\vec{\tau}_{net}$ is the sum of $\tau_1, \tau_2, \tau_3 \dots \tau_n$ then:-

$$\tau_{net} = \tau_1 + \tau_2 + \tau_3 + \dots \tau_n = 0$$

$$\Sigma \tau = 0$$

Given :-

f_1 :- 90N

d_1 : 3.6m

F_2 : 210N

d_2 : 1.2m. support : 1.2m

f : ?

Centre of plank :-

$\frac{3.6}{2} = 1.8m$

distance :- $1.8 - 1.2$
$= 0.6m$

distance of boy from support :-
1.2m

Distance of F from support :-

$3.6 - 1.2 = 2.4m$

Principle of moment :-

Clockwise moment = Anticlockwise moment.

$210 \times 1.2 + 90 \times 0.6 = F(2.4)$

$252 + 54 = 2.4F$

$306 = 2.4F$

$F = \frac{306}{2.4} \Rightarrow 127.5\,N$

— Q:3(ii) —

Power is the rate at which work is done or energy is transfered.

formula :- $P = \frac{W}{t}$

SI unit : Watt (W)

Given :-

$m = 850kg$
$h = 24m$
$t = 40s$
$g = 10m/s^2$

(i) Potential energy :- mgh

$850 \times 10 \times 24$

$204,000 J$

So the girder gains 240 kJ of PE.

(ii) Power output :-

$P = \frac{\text{energy transfered}}{\text{time}}$

$= \frac{204.000}{40}$

$= 5100 W$

Watt into Kilowatts:-

$$\frac{5100}{1000} = 5.1 \text{ kW}$$

(c) Electrical energy → Kinetic/ mechanical energy.